Interval Observers

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Motivation

Motivation of this approach

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* We consider a class of **"uncertain"** dynamical systems:

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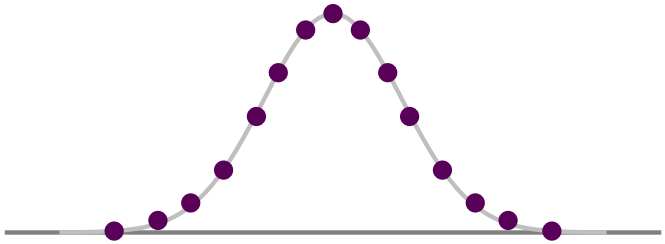
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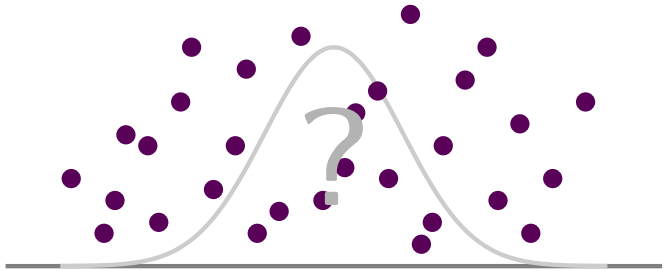
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* The uncertianty is on the the initial condition or in the parameters of the system.
* One big question is: how do we measure the uncertainty on our model?
  - **Parametric approach:** Assumes a distribution on parameters — often hard to verify in biology.
  - **Functional approach:** Assumes bounded uncertainty — uses interval observers.

Statistical Assumption



Biological Reality



Definitions

Recall our system:

$$(S_0): \begin{cases} \dot{x}(t) = f(t, x(t)), \\ y(t) = h(t, x(t)) \end{cases} \qquad x(0)=x_0$$

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- Consider a family of systems parameterized by $K \in \mathcal{K}$:

$$(\hat{S}_K): \begin{cases} \dot{z}(t) = g_K\left(t, z(t), y(t)\right), \\ \hat{x}(t) = \ell_K\left(t, z(t), y(t)\right) \end{cases} \qquad z(0)=z_0$$

where $z(t) \in \mathcal{Z} \subset \mathbb{R}^m$ is the internal state.

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- **Definition 3 (Expo obs).** $(\hat{S}_K)$ is an exponential observer if:

$\forall \lambda > 0$, $\exists K_\lambda$ s.t. $(S_0, \hat{S}_{K_\lambda})$ satisfies:

$$\forall (x_0, z_0) \in \mathcal{X} \times \mathcal{Z}, \ \forall t \geq 0,$$
$$\|\hat{x}(t) - x(t)\| \leq e^{-\lambda t} \|\hat{x}(0) - x(0)\|.$$

The Linear Case

Consider the linear autonomous system:

$$(S_L): \begin{cases} \dot{x}(t) = Ax(t), \\ y(t) = Cx(t) \end{cases}$$

with $A \in \mathcal{M}_{n\times n}(\mathbb{R})$ $(n \geq 2)$,
$C \in \mathcal{M}_{1\times n}(\mathbb{R})$.

Observability Criterion (Kailath, 1980):

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Luenberger Observer.
If (S_L) is observable, there exists an **exponential observer** in Luenberger form:

$$\frac{d\hat{x}(t)}{dt} = A\hat{x}(t) + K\left(C\hat{x}(t) - y(t)\right)$$

where $K \in \mathbb{R}^{n\times 1}$ is a gain vector chosen such that $A + KC$ has all its eigenvalues negative real parts.

Error dynamics:

$$\dot{e}(t) = (A + KC)e(t)$$

with $e(t) = \hat{x}(t) - x(t) \to 0$ exponentially.

The Affine Output Injection Feedback Case

Consider the system (S):

$$(S): \begin{cases} \dot{x}(t) = Ax(t) + f(t, y(t)), \\ y(t) = Cx(t) \end{cases}$$

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The nonlinearity $f : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}^n$ depends **only on the output** and is assumed to be:

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Observer Design.
If the pair (A, C) is observable and f is known, one can design the observer:

$$\frac{d\hat{x}(t)}{dt} = A\hat{x}(t) + f(t, y(t)) + K\big(C\hat{x}(t) - y(t)\big)$$

Error dynamics:

$$\dot{e}(t) = (A + KC)e(t)$$

Same structure as the linear case!

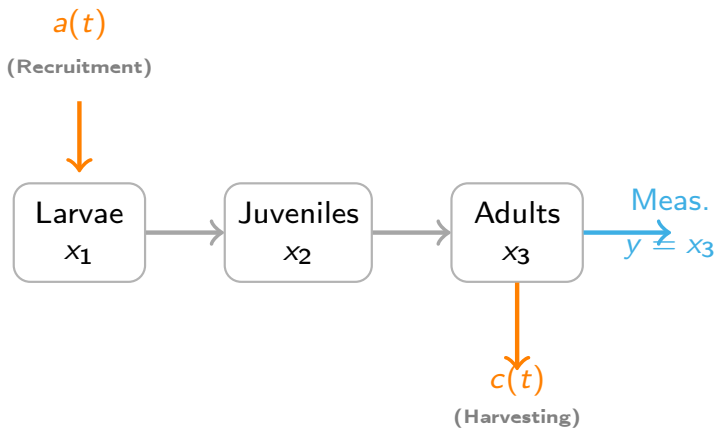
Fish population model

Estimating unmeasured stocks in a fish population model

Case Study 1: Fish population model

Consider a population structured into three stages: larvae, juveniles and adults whose respective stocks $x = (x_1, x_2, x_3) \in \mathcal{X} = \mathbb{R}^3_+$ follow the dynamics:

$$\begin{cases} \dot{x}_1 = -a_1 x_1 - m_1 x_1 + r(t, x_3) \\ \dot{x}_2 = a_1 x_1 - a_2 x_2 - m_2 x_2 \\ \dot{x}_3 = a_2 x_2 - m_3 x_3 - c(t) x_3 \\ y = x_3 \end{cases}$$



In the form $\dot{x}(t) = Ax(t) + f(t, y(t))$ with $a(t) > 0$, $b > 0$:

$$A = \begin{pmatrix} -(a_1 + m_1) & 0 & 0 \\ a_1 & -(a_2 + m_2) & 0 \\ 0 & a_2 & -m_3 \end{pmatrix}, \quad f(t, y) = \begin{pmatrix} \dfrac{a(t)y}{b + y} \\ 0 \\ -c(t)y \end{pmatrix}$$

where $f : \mathbb{R}_+ \times \mathbb{R} \to \mathbb{R}^3$ is **measurable** in t and **Lipschitz continuous** in y (in each coordinate).

Definitions: Key Ideas

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- **The spirit of this approach:**

  Given bounds on the uncertainties in the model, we are looking for **dynamic bounds** on the estimation of the variables.

- Because bounds are usually given by **intervals**, one shall build what is called **"interval observers"**.

What is an Interval Observer?

A pair of systems (S^-, S^+)**:**

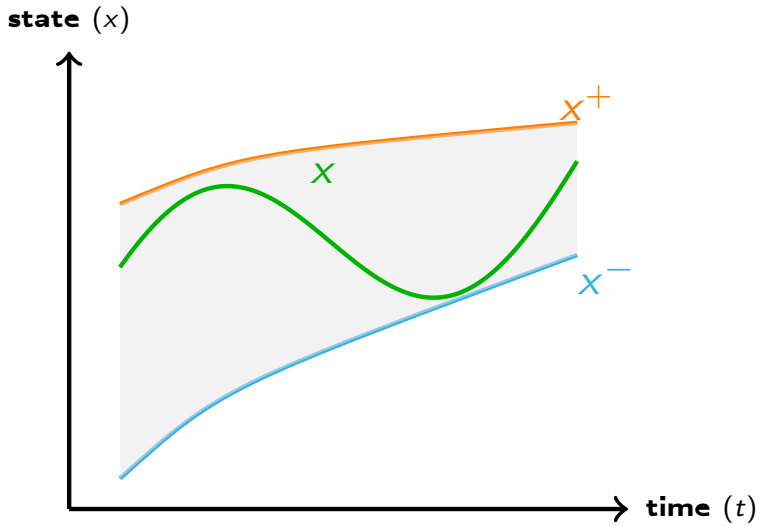
$$(S^-): \begin{cases} \dot{z}^-(t) = g^-\big(t, z^-(t), y(t)\big), \\ x^-(t) = \ell^-\big(t, z^-(t), z^+(t), y(t)\big) \end{cases}$$

$$(S^+): \begin{cases} \dot{z}^+(t) = g^+\big(t, z^+(t), y(t)\big), \\ x^+(t) = \ell^+\big(t, z^-(t), z^+(t), y(t)\big) \end{cases}$$

is an **interval observer** for the system (S_0) if for any compact set $X_0 \subset \mathcal{X}$, there exist initial conditions $(z^-(0), z^+(0))$ such that (S_0, S^-, S^+) verifies:

$$x^-(t) \leq x(t) \leq x^+(t), \ \forall t \geq 0.$$

Interpretation: The trajectories $x^-(t)$ and $x^+(t)$ form a tube that contains the true state trajectory $x(t)$ for all times $t \geq 0$.



Construction for Almost-Linear Systems

Assumption: For system (2), there exist known applications $f^-, f^+ : \mathbb{R}^+ \times \mathbb{R}^n \to \mathbb{R}^n$, measurable w.r.t. t and Lipschitz continuous w.r.t. y such that:

$$f^-(t,y) \leq f(t,y) \leq f^+(t,y),$$

$\forall (t,y) \in \mathbb{R}^+ \times \mathbb{R}^n$

Upper estimate x^+:

$$\dot{x}^+(t) = Ax^+(t) + f^+(t,y(t)) + K(Cx^+(t) - y(t))$$

Lower estimate x^-:

$$\dot{x}^-(t) = Ax^-(t) + f^-(t,y(t)) + K(Cx^-(t) - y(t))$$

Error analysis: For the upper error $e^+(t) = x^+(t) - x(t)$:

$$\dot{e}^+ = (A + KC)e^+ + b^+(t)$$

where

$$b^+(t) = f^+(t,y(t)) - f(t,y(t)) \geq 0$$

The nonnegativity of $b^+(t)$ follows from the bounds on f. This enables application of **cooperative systems** theory.

Monotone dynamic System and Cooperative

Functional Framing